

Dr. Mahdi AbuAli, PhD

GDAC Saudi Lead



GDAC-SA Tender Role: Senior Technical Advisor | 35+ Years Saudi Aramco | Fleet Tahreez Managing Director (KSA) | 50+ Prospects Risked | Native Arabic Speaker

Senior Geological Consultant - Saudi Arabia Petroleum Expert

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Current Positions

COO and VP

Raphael Energy Group | Australia (Remote / Australia & KSA) *January 2021 - Present (5 years) | Part-time*

Provide corporate leadership across portfolio strategy, governance cadence, and AOP/budget ownership, working with directors and partners to align technical evaluation with investment decisions. Focused on operating discipline, HSE/ESG, and building digital capability.

Key Achievements: - Established board reporting and rhythm for AOP, performance, and governance - Integrated 3D grav/mag with IHI to prioritize opportunities and move prospects through internal stage-gates - Strengthened stakeholder engagement with directors, investors, and partners

Raphael Energy Group Limited is exploring for oil and gas in the Officer Basin, onshore Western Australia, using an innovative approach integrating 3D gravity/magnetic mapping with microseepage induced indirect hydrocarbon indications (IHIs) such as radiometric gamma anomalies, to provide 3D GravMag IHI leads.

Managing Director

Fleet Tahreez Company Limited | Riyadh, Saudi Arabia *July 2025 - November 2025 (5 months) | Full-time*

Leading mineral exploration initiatives that apply satellite- and AI-enabled geoscience to generate, rank, and de-risk targets for tier-one discoveries. Bridge operator needs with advanced sensing, turning technical insight into decision inputs for work-program planning and stakeholder reviews across KSA.

Key Achievements: - Introduced satellite/AI workflows that improved speed of exploration, target generation and prioritization - Delivered decision inputs adopted in planning reviews across KSA

Senior Advisor

Axiom Exploration Group | Saskatchewan, Canada (Hybrid) *April 2021 - June 2025 (4 years 3 months) | Part-time*

Expanded remit from hydrocarbons to minerals, reflecting demand and the energy transition.

Key Achievements: - Supported opportunity screening and BD engagements for regional mandates - Broadened scope to minerals exploration aligned with market priorities

Recent Consulting Experience

Consultant & Project Manager

Halliburton Kuwait | Kuwait *February 2023 - October 2023 (9 months) | Full-time*

Led a multidisciplinary team on Paleozoic/Triassic gas exploration, delivering a ranked prospect inventory with full risking and volumetrics to inform client decision-making and program planning.

Key Achievements: - Matured, ranked, and risked 10 prospects across ~3,000 km² with supporting volumetrics - Inventory adopted for the next stage of planning by KOC stakeholders

Saudi Aramco - 35 Years 9 Months of Distinguished Service

April 1985 - December 2020

Senior Geological Consultant

Saudi Aramco - Upstream Advanced Research Center | Dhahran, Eastern, Saudi Arabia June 2014 - December 2020 (6 years 7 months) | Full-time

Drove deployment of exploration technologies to lower risk, mentored junior geoscientists, and embedded new methods into daily workflows—strengthening technical quality and decision confidence. Published papers and authored patents.

Key Achievements: - Embedded risk-reducing technologies and practices across teams - Delivered internal courses and editorial leadership to raise technical standards

Senior Geological Consultant

Saudi Aramco - Strategic Planning & Optimization (Exploration) | Dhahran, Saudi Arabia February 2011 - June 2014 (3 years 5 months) | Full-time

Member of a senior team standardizing portfolio risking and shaping the exploration drilling schedule. Partnered with leadership on investment cases, approvals, and AOP delivery.

Key Achievements: - Standardized prospect risking across the national portfolio; matured and risked 50+ prospects, several drilled and resulting in oil & gas discoveries - Prepared decision papers and presented to senior leadership to inform approvals and execution

Senior Geological Consultant

Saudi Aramco - Upstream Ventures (JV Programme Manager) | Dhahran, Saudi Arabia September 2006 - February 2011 (4 years 6 months) | Full-time

Managed JV exploration programs (SRAK, EniRepSa, SSG, LukSar), representing Aramco in governance forums and partner committees. Oversight spanned large onshore areas with planned wells and development concepts progressed with JV partners.

Key Achievements: - Oversaw >200,000 km² of JV acreage; planned several exploration wells and advanced the Kidan Arab sour-gas development plan - Prepared decision papers and secured partner approvals for work programs and budgets

Senior Geologist - Earlier Roles

Saudi Aramco | Dhahran, Saudi Arabia June 1989 - September 2006 (17 years 4 months) | Full-time

Progression: Geochemist → Reservoir/Exploration Geologist → Geological Specialist/Team Lead

Progressed from hands-on geochemistry into exploration and specialist leadership, supporting prospect generation through petroleum systems modeling, source–oil correlations, reservoir characterization (clastic & carbonate), and 2D/3D seismic interpretation. Partnered with multi-disciplinary teams to translate subsurface insight into drillable opportunities and helped establish repeatable workflows adopted across exploration.

Key Achievements: - Built regional play-fairway and charge/migration frameworks used in exploration interpretation - Delivered geochemical correlations and subsurface interpretations that underpinned prospect maturation and investment decisions

Geologist

Saudi Aramco | Dhahran, Saudi Arabia April 1985 - September 1996 (11 years 6 months) | Full-time

Geologist and Geochemist, worked on Exploration of Central Arabia, Northwest and Red Sea areas. Conducted R&D studies on the source rock and migration of hydrocarbons, established a new source rock for the Paleozoic of Saudi Arabia. Published and presented many articles on the origin of hydrocarbons in the Paleozoic petroleum systems.

Core Expertise

- **Regional Geology:** Deep expertise in Arabian Peninsula, Middle East petroleum systems
- **Exploration Strategy:** Portfolio management, prospect risking, resource assessment
- **Basin Analysis:** Petroleum systems modeling, play concept development
- **Leadership:** Team building, mentoring, stakeholder management, multicultural team leadership
- **Saudi Operations:** 35+ years working in Saudi Arabia with comprehensive understanding of local context
- **Technical Integration:** Multidisciplinary data integration and interpretation
- **Business Development:** Partnership negotiations, JV management, joint ventures
- **Emerging Technologies:** Satellite/AI-enabled exploration, 3D GravMag IHI methods
- **Minerals Exploration:** Recent experience expanding from hydrocarbons to critical minerals

Technical Skills

- **Exploration Management:** Project management, exploration portfolio analysis, budgeting, strategic planning
- **Geological Analysis:** Basin analysis, petroleum systems analysis, reservoir characterization, field work, sampling
- **Data & Analytics:** Data analysis, quantitative research, qualitative research, analytical skills
- **Technical Disciplines:** Seismic interpretation, log interpretation, geochemistry, source rock analysis
- **Leadership & Management:** Team management, coaching, multicultural team leadership, contractual obligations
- **Specialized:** AI exploration, satellite-enabled exploration, mineral exploration, HSE/ESG

Education

PhD - Geosciences

RWTH Aachen University | Germany *2001 - 2005*

- Advanced research in petroleum systems and basin analysis
- Published research in peer-reviewed journals
- Activities and societies: AAPG, SPE, EAGE, IMOG, ALAGO, AAPG

Master of Science (MS) - Geochemistry

Colorado School of Mines | Colorado, USA *January 1986 - May 1989*

- **Grade: 3.2/4.0**
- Specialization in petroleum geochemistry

Bachelor of Applied Science (BASc) - Industrial Chemistry

King Fahd University of Petroleum & Minerals (KFUPM) | Dhahran, Saudi Arabia *September 1980 - January 1985*

- **Grade: 3.0/4.0**
- Strong foundation in chemistry and petroleum sciences

Languages

- **Arabic:** Native speaker - critical for Saudi stakeholder engagement
- **English:** Fluent - international business and technical communication

Professional Memberships

- American Association of Petroleum Geologists (AAPG)
- Society of Petroleum Engineers (SPE)
- European Association of Geoscientists and Engineers (EAGE)

Relevance to GDAC-SA Tender

- **Saudi Arabia Expertise:** 35+ years 9 months working in Saudi Arabia at Saudi Aramco (1985-2020) - unparalleled understanding of local geology, culture, and stakeholder landscape
- **Recent Saudi Experience:** Managing Director at Fleet Tahreez (Jul-Nov 2025) in Riyadh - current, active relationships with Saudi geological authorities and stakeholders
- **Arabic Language:** Native speaker enables direct, nuanced communication with Saudi Geological Survey and government stakeholders
- **Senior Advisory Capability:** Current COO/VP at Raphael Energy and recent Senior Advisor roles demonstrate continued relevance and thought leadership
- **Proven Leadership:** Led major strategic initiatives at Saudi Aramco managing >200,000 km² and risking 50+ prospects with multiple discoveries
- **Resource Assessment:** Deep expertise in probabilistic risking and resource estimation - directly applicable to GDAC database requirements
- **International Collaboration:** Extensive JV experience with major oil companies (ENI, Lukoil, Shell) - knows how to work with diverse stakeholders and partners
- **Data Integration:** Career focused on integrating multidisciplinary geoscience data - core requirement for GDAC platform development
- **Mentoring:** Decades of experience training and developing geoscientists - can support knowledge transfer to Saudi teams
- **Gulf Region Network:** Extensive professional network across Saudi Arabia and broader Gulf region from 35+ years of service
- **Innovation Focus:** Recent experience with satellite/AI-enabled exploration and 3D GravMag IHI methods demonstrates comfort with emerging technologies
- **Minerals Expertise:** Expanded scope to minerals exploration aligns with Saudi Vision 2030 diversification beyond hydrocarbons
- **Current Saudi Presence:** Recent 5-month assignment in Riyadh (2025) demonstrates ability to work on-ground in KSA